

NETWORK TRAFFIC ANALYSIS

Wireshark PCAP Forensics Report

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Course: Ethical Hacking

Assignment: Wireshark PCAP Analysis (20 Questions)

File Analyzed: traffic.pcapng

Tool: Wireshark

Objective

This report presents a forensic analysis of the provided network packet capture file, traffic.pcapng, using Wireshark. The analysis focuses on extracting and verifying twenty key data points related to network communication, including protocol statistics, packet counts, IP and MAC address mappings, host communication patterns, vendor identification, and overall traffic characteristics.

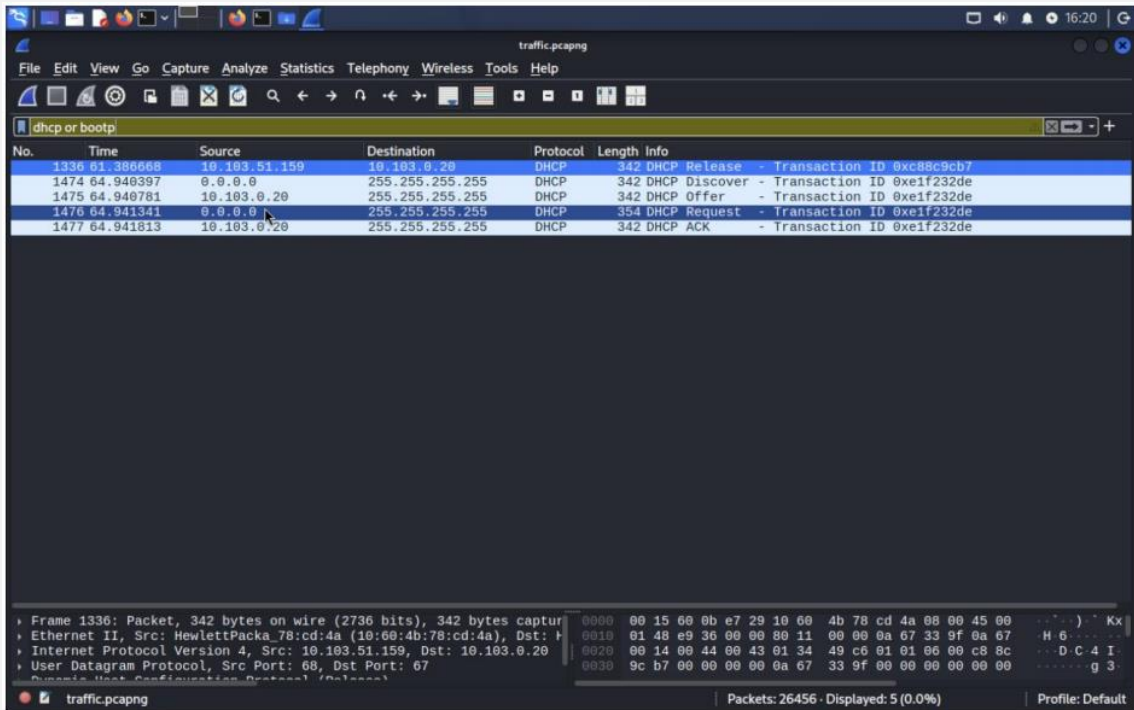
Capture File Overview

The following section presents a detailed analysis of each assignment question. Every answer below was verified against the raw packet headers of traffic.pcapng .

Detailed Question-by-Question Analysis

E1. Download and analyze the network file; find the number of DHCP messages.

Ans :- 5



The screenshot shows the Wireshark interface with the filter 'dhcp or bootp' applied. The packet list displays five DHCP messages:

No.	Time	Source	Destination	Protocol	Length	Info
1336	64.386668	10.103.51.159	10.103.0.20	DHCP	342	DHCP Release - Transaction ID 0xc88c9cb7
1474	64.940397	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe1f232de
1475	64.940781	10.103.0.20	255.255.255.255	DHCP	342	DHCP Offer - Transaction ID 0xe1f232de
1476	64.941341	0.0.0.0	255.255.255.255	DHCP	354	DHCP Request - Transaction ID 0xe1f232de
1477	64.941813	10.103.0.20	255.255.255.255	DHCP	342	DHCP ACK - Transaction ID 0xe1f232de

Below the packet list, the details of the selected packet (Frame 1336) are shown, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol fields.

Filter used: `udp.port == 67 || udp.port == 68` (or: bootp)

E2. Analyze the network file and find the number of ARP messages.

Ans :- 680 packets

No.	Time	Source	Destination	Protocol	Length	Info
1260	58.250648	HewlettPacka_a6:42::	Broadcast	ARP	60	who has 10.103.50.88? Tell 10.103.50.249
1263	58.375596	HewlettPacka_9e:5b::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.221
1265	58.420696	Intel_6c:26:0a	Broadcast	ARP	60	who has 10.103.200.5? Tell 10.103.0.18
1266	58.435875	Intel_6c:26:0a	Broadcast	ARP	60	who has 10.103.51.15? Tell 10.103.0.18
1268	58.708373	HewlettPacka_9b:74::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.51.13
1274	58.950524	Intel_6c:26:0a	Broadcast	ARP	60	who has 10.103.200.5? Tell 10.103.0.18
1280	59.130577	HewlettPacka_a6:42::	Broadcast	ARP	60	who has 10.103.50.88? Tell 10.103.50.249
1282	59.258881	HewlettPacka_78:cd::	Broadcast	ARP	42	who has 10.103.0.20? Tell 10.103.51.159
1283	59.259022	HewlettPacka_0b:e7::	Broadcast	ARP	60	who has 10.103.0.20 is at 00:15:00:0b:e7:29
1288	59.367507	HewlettPacka_9e:5b::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.221
1292	59.710760	HewlettPacka_9b:74::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.51.13
1296	59.849020	Intel_6c:26:0a	Broadcast	ARP	60	who has 10.103.200.5? Tell 10.103.0.18
1297	59.972945	HewlettPacka_9e:5d::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.51.43
1301	60.137612	HewlettPacka_a6:42::	Broadcast	ARP	60	who has 10.103.50.88? Tell 10.103.50.249
1322	60.559142	HewlettPacka_9b:74::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.190
1323	60.693429	HewlettPacka_9e:5d::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.51.43
1332	61.190831	HewlettPacka_9b:73::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.8
1333	61.257843	HewlettPacka_9b:74::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.190
1340	61.592108	HewlettPacka_9e:5b::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.221
1353	61.692500	HewlettPacka_9e:5d::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.51.43
1354	61.751009	HewlettPacka_9b:73::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.8
1366	62.257876	HewlettPacka_9b:74::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.190
1369	62.375702	HewlettPacka_9e:5b::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.221
1397	62.756284	HewlettPacka_9b:73::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.8
1414	63.376310	HewlettPacka_9e:5b::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.221
1419	63.502677	BrotherIndus_93:d4::	Broadcast	ARP	60	who has 10.103.50.37? Tell 10.103.200.10
1430	63.834024	Intel_6c:26:0a	Broadcast	ARP	60	who has 10.103.200.5? Tell 10.103.0.18
1444	64.007075	HewlettPacka_a6:42::	Broadcast	ARP	60	who has 10.103.50.88? Tell 10.103.50.249
1450	64.235454	HewlettPacka_9b:73::	Broadcast	ARP	60	who has 10.103.51.22? Tell 10.103.50.8

Frame 1333: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
 Ethernet II, Src: HewlettPacka_9b:74:f7 (a0:48:1c:9b:74:f7), Dst: Broadcast
 Address Resolution Protocol (request)
 0000 ff ff ff ff ff ff a0 48 1c 9b 74 f7 08 06 00 00
 0010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 0020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
 0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Filter used: arp

E3. Analyze the network file and find the IP address that accessed 'baidu.com'.

Ans :- 10.103.51.159

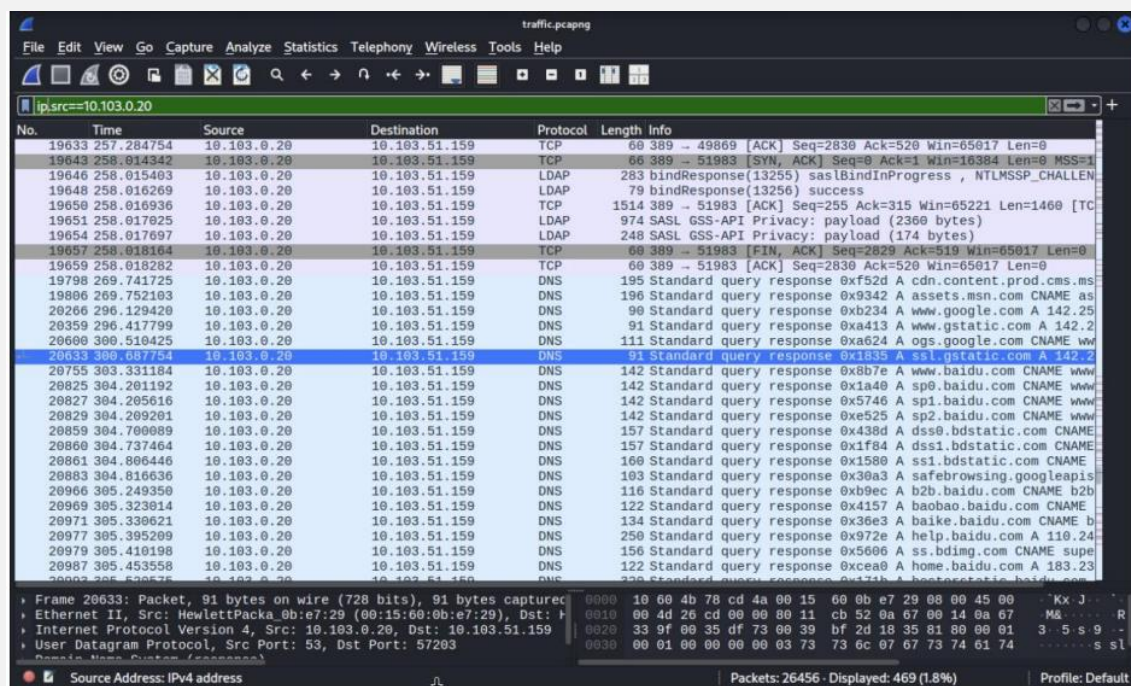
No.	Time	Source	Destination	Protocol	Length	Info
20754	393.328979	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x8b7e A www.baidu.com
20755	393.331184	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x8b7e A www.baidu.com CNAME www
20823	394.130182	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x1a40 A sp0.baidu.com
20825	394.201192	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x1a40 A sp0.baidu.com CNAME www
20826	394.203706	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x5746 A sp1.baidu.com
20827	394.205616	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0x5746 A sp1.baidu.com CNAME www
20828	394.207066	10.103.51.159	10.103.0.20	DNS	73	Standard query 0xe525 A sp2.baidu.com
20829	394.209201	10.103.0.20	10.103.51.159	DNS	142	Standard query response 0xe525 A sp2.baidu.com CNAME www
20894	394.880525	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x36e3 A baike.baidu.com
20895	394.880525	10.103.51.159	10.103.0.20	DNS	73	Standard query 0xb9ec A b2b.baidu.com
20896	394.880553	10.103.51.159	10.103.0.20	DNS	76	Standard query 0x4157 A baobao.baidu.com
20897	394.884088	10.103.51.159	10.103.0.20	DNS	82	Standard query 0x171b A hectorstatic.baidu.com
20966	395.249350	10.103.0.20	10.103.51.159	DNS	116	Standard query response 0xb9ec A b2b.baidu.com CNAME b2b
20969	395.323914	10.103.0.20	10.103.51.159	DNS	122	Standard query response 0x4157 A baobao.baidu.com CNAME
20970	395.324575	10.103.51.159	10.103.0.20	DNS	76	Standard query 0x5973 A haokan.baidu.com
20971	395.330621	10.103.0.20	10.103.51.159	DNS	134	Standard query response 0x36e3 A baike.baidu.com CNAME b
20972	395.331623	10.103.51.159	10.103.0.20	DNS	74	Standard query 0x972e A help.baidu.com
20977	395.395209	10.103.0.20	10.103.51.159	DNS	250	Standard query response 0x972e A help.baidu.com A 110.24
20978	395.396834	10.103.51.159	10.103.0.20	DNS	74	Standard query 0xcea0 A home.baidu.com
20987	395.453558	10.103.0.20	10.103.51.159	DNS	122	Standard query response 0xcea0 A home.baidu.com A 183.23
20989	395.454811	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x98fa A image.baidu.com
20993	395.520575	10.103.0.20	10.103.51.159	DNS	320	Standard query response 0x171b A hectorstatic.baidu.com
20997	395.534356	10.103.51.159	10.103.0.20	DNS	72	Standard query 0xa5aa A ir.baidu.com
21036	395.660962	10.103.0.20	10.103.51.159	DNS	162	Standard query response 0xa5aa A ir.baidu.com CNAME ir.b
21037	395.662353	10.103.51.159	10.103.0.20	DNS	77	Standard query 0xb46a A jingyan.baidu.com
21064	395.810768	10.103.0.20	10.103.51.159	DNS	151	Standard query response 0x5973 A haokan.baidu.com CNAME
21065	395.812277	10.103.51.159	10.103.0.20	DNS	74	Standard query 0xd2c0 A live.baidu.com
21066	395.868327	10.103.0.20	10.103.51.159	DNS	134	Standard query response 0xd2c0 A live.baidu.com CNAME po
21067	395.869886	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x28a3 A map.baidu.com

Frame 20754: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface 0
 Ethernet II, Src: HewlettPacka_78:cd:4a (10:60:4b:78:cd:4a), Dst: 10.103.0.20
 Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20
 User Datagram Protocol, Src Port: 62733, Dst Port: 53
 Domain Name System (query)
 0000 00 15 00 0b e7 29 10 00 4b 78 cd 4a 08 00 45 00
 0010 00 2b 4e 0f 00 00 00 11 00 00 0a 07 33 9f 0a 67
 0020 00 14 f5 0d 00 35 00 27 48 b9 8b 7e 01 00 00 01
 0030 00 00 00 00 00 00 03 77 77 77 05 02 61 69 64 75

Filter used: dns.qry.name contains "baidu"

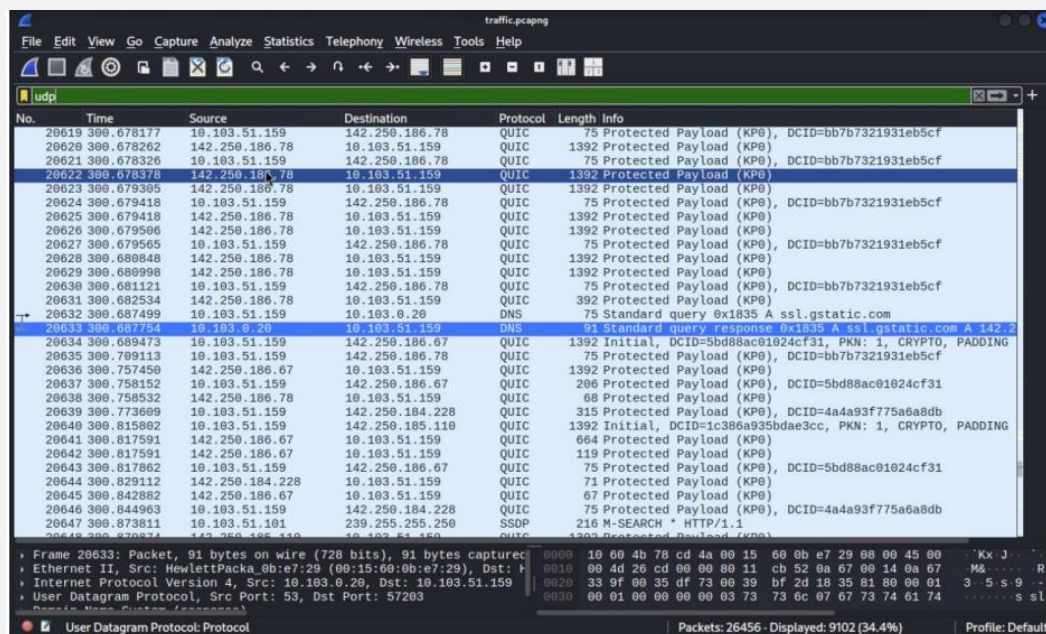
E4. Analyze the network file and find the number of packets the IP address 10.103.0.20 sent.

Ans :- 509 packets



E5. Analyze the network file and find the number of UDP packets.

Ans :- 8,610 packets



E6. Analyze the network file and find the number of SMB packets.

Ans :- 518 packets

No.	Time	Source	Destination	Protocol	Length	Info
26139	287.477687	10.103.50.245	10.103.255.255	BROWSER	243	Host Announcement 489_04, Workstation, Server, SQL Serve
26177	288.561175	10.103.51.100	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26193	289.212617	10.103.50.80	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26230	291.547787	10.103.50.82	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26233	292.024916	10.103.50.208	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26240	293.145826	10.103.51.19	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26250	293.975914	10.103.51.138	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26254	294.485368	10.103.50.83	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26263	295.997346	10.103.51.141	10.103.255.255	BROWSER	243	Host Announcement 489_08, Workstation, Server, SQL Serve
26398	296.783177	10.103.50.194	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26408	297.277775	10.103.51.138	10.103.255.255	BROWSER	243	Host Announcement 489_01, Workstation, Server, SQL Serve
26527	298.746841	10.103.50.86	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26589	299.707839	10.103.50.5	10.103.255.255	BROWSER	243	Host Announcement LAPTOP-4P07CS85, Workstation, Server,
26592	299.929461	10.103.50.184	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26585	300.156789	10.103.50.163	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26744	302.686797	10.103.50.129	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26747	303.043967	10.103.51.142	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26749	303.139374	10.103.51.107	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26752	303.256316	10.103.50.85	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26759	303.376889	10.103.50.41	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26764	303.666957	10.103.51.109	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26771	303.866085	10.103.51.141	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
26915	305.105681	10.103.50.39	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21062	305.784435	10.103.51.1	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21190	306.478651	10.103.51.20	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21192	306.498811	10.103.51.39	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21217	306.885479	10.103.50.233	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21227	307.072882	10.103.50.8	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21234	307.353686	10.103.50.37	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]
21235	307.370104	10.103.51.12	10.103.255.255	SMB_NE...	280	SAM LOGON request from client[Malformed Packet]

Frame 26595: Packet, 280 bytes on wire (2240 bits), 280 bytes captured on interface 0, from 10.103.50.163 to 10.103.255.255 on interface 0
 Ethernet II, Src: Hewlett-Packard, 9e:ec:0c:78:ac:c0:9e:ec:0c, Dst: 01:00:00:00:00:00
 Internet Protocol Version 4, Src: 10.103.50.163, Dst: 10.103.255.255
 User Datagram Protocol, Src Port: 138, Dst Port: 138
 SMB2 (Server Message Block Protocol version 2): Protocol

Filter used: smb || smb2

E7. Analyze the network file and find the number of packets sent to the source IP address 10.103.0.20.

Ans :- 562 packets

No.	Time	Source	Destination	Protocol	Length	Info
19632	257.284676	10.103.51.159	10.103.0.20	TCP	54	49869 → 389 [ACK] Seq=520 Ack=2830 Win=2102016 Len=0
19642	258.014179	10.103.51.159	10.103.0.20	TCP	60	51983 → 389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
19644	258.014408	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [ACK] Seq=1 Ack=1 Win=2102272 Len=0
19645	258.015047	10.103.51.159	10.103.0.20	LDAP	135	bindRequest(13255) "c-root", NTLMSSP_AUTH, User: \sas
19647	258.015725	10.103.51.159	10.103.0.20	LDAP	214	bindRequest(13256) "c-root", NTLMSSP_AUTH, User: \sas
19649	258.016384	10.103.51.159	10.103.0.20	LDAP	127	SASL GSS-API Privacy: payload (53 bytes)
19652	258.017046	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [ACK] Seq=315 Ack=2635 Win=2102272 Len=0
19653	258.017417	10.103.51.159	10.103.0.20	LDAP	226	SASL GSS-API Privacy: payload (152 bytes)
19655	258.017938	10.103.51.159	10.103.0.20	LDAP	86	SASL GSS-API Privacy: payload (12 bytes)
19656	258.018086	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [FIN, ACK] Seq=519 Ack=2829 Win=2102016 Len=0
19658	258.018197	10.103.51.159	10.103.0.20	TCP	54	51983 → 389 [ACK] Seq=520 Ack=2830 Win=2102016 Len=0
19797	269.741416	10.103.51.159	10.103.0.20	DNS	88	Standard query 0xf52d A cdn.content.prod.cms.msn.com
19805	269.751901	10.103.51.159	10.103.0.20	DNS	74	Standard query 0x9342 A assets.msn.com
20265	296.129207	10.103.51.159	10.103.0.20	DNS	74	Standard query 0xb234 A www.google.com
20358	296.416562	10.103.51.159	10.103.0.20	DNS	75	Standard query 0xa413 A www.gstatic.com
20599	300.508422	10.103.51.159	10.103.0.20	DNS	74	Standard query 0xa624 A ogs.google.com
20632	300.687499	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x1835 A ssl.gstatic.com
20754	303.328879	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x8b7e A www.baidu.com
20821	304.130181	10.103.51.159	10.103.0.20	DNS	77	Standard query 0x438d A dss.bdstatic.com
20822	304.130182	10.103.51.159	10.103.0.20	DNS	77	Standard query 0x1f84 A dss.bdstatic.com
20823	304.130182	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x1a40 A sp0.baidu.com
20826	304.203766	10.103.51.159	10.103.0.20	DNS	73	Standard query 0x5746 A sp1.baidu.com
20828	304.207066	10.103.51.159	10.103.0.20	DNS	73	Standard query 0xe525 A sp2.baidu.com
20830	304.210110	10.103.51.159	10.103.0.20	DNS	76	Standard query 0x1580 A ssl.bdstatic.com
20881	304.813911	10.103.51.159	10.103.0.20	DNS	72	Standard query 0x5606 A ss.bdimg.com
20882	304.816438	10.103.51.159	10.103.0.20	DNS	87	Standard query 0x30a3 A safebrowsing.googleapis.com
20894	304.880525	10.103.51.159	10.103.0.20	DNS	75	Standard query 0x36e3 A baibe.baidu.com
20895	304.880525	10.103.51.159	10.103.0.20	DNS	73	Standard query 0xb9ec A b2b.baidu.com
20896	304.880525	10.103.51.159	10.103.0.20	DNS	76	Standard query 0x4157 A baobao.baidu.com
20897	304.880525	10.103.51.159	10.103.0.20	DNS	82	Standard query 0x474b A baobao.baidu.com

Frame 20358: Packet, 75 bytes on wire (600 bits), 75 bytes captured on interface 0, from 10.103.51.159 to 10.103.0.20 on interface 0
 Ethernet II, Src: Hewlett-Packard, 78:cd:4a:10:60:4b:78:cd:4a, Dst: 08:00:00:00:00:00
 Internet Protocol Version 4, Src: 10.103.51.159, Dst: 10.103.0.20
 User Datagram Protocol, Src Port: 61980, Dst Port: 53

Filter used: ip.dst == 10.103.0.20

E8. Examine the network file and isolate only the IPv4 traffic; determine the count of packets shown.

Ans :- 24,906 packets

Filter used: ip

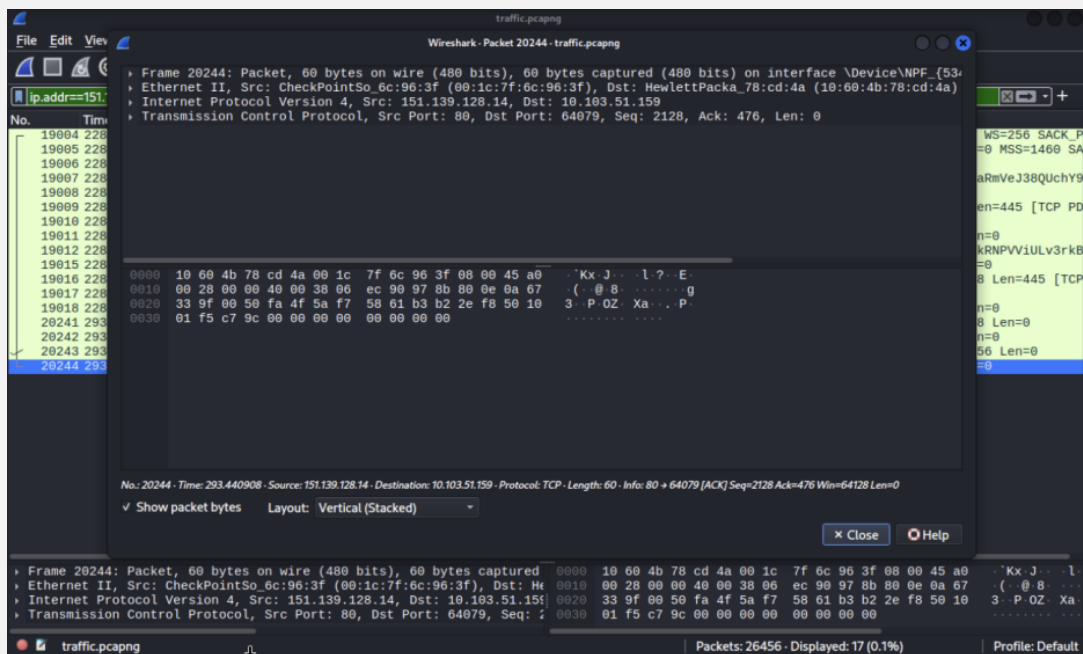
E9. Analyze the network file and examine the last SMB packet; identify the source IP address.

Ans :- 10.103.50.202

Filter used: smb || smb2

E10. Analyze the network file and find the MAC address of the IP 151.139.128.14.

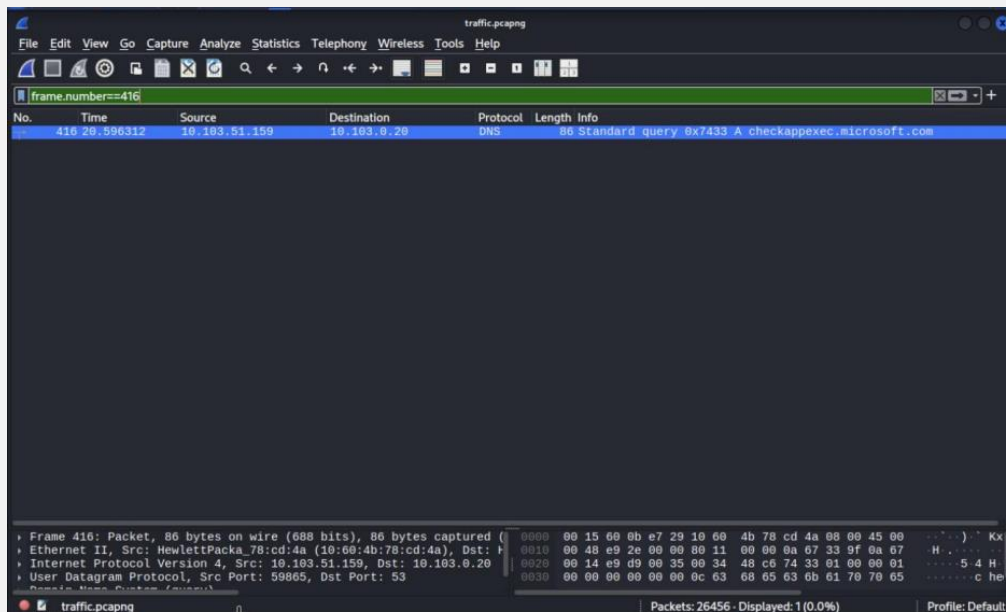
Ans :- 00:1c:7f:6c:96:3f



Filter used: ip.addr == 151.139.128.14

E11. Examine packet 416 in the network file and identify the protocol being used.

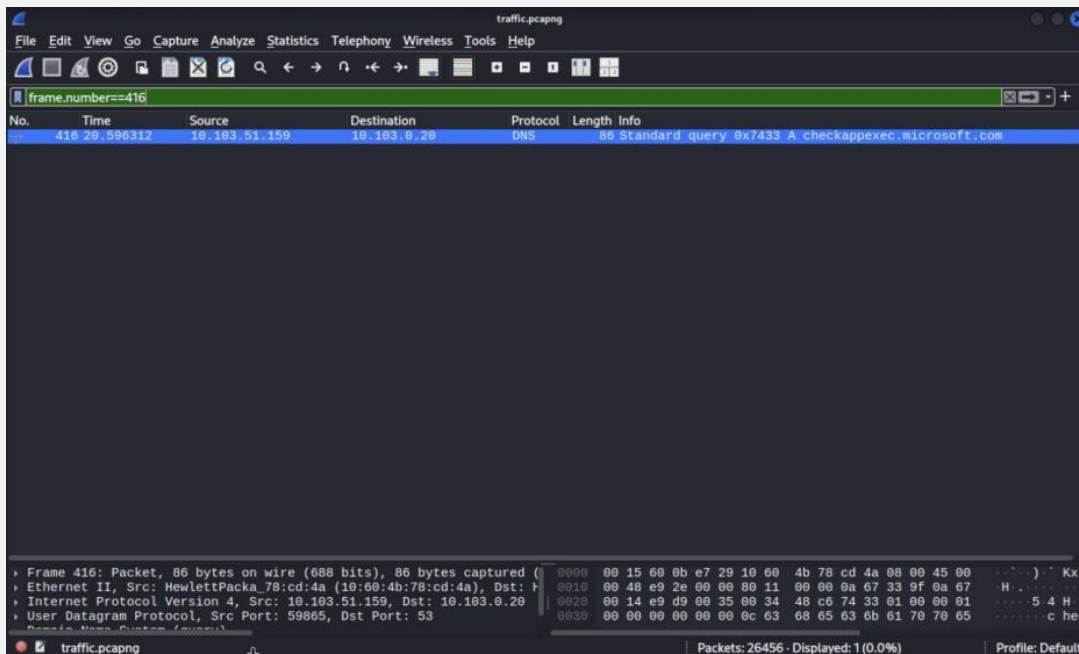
Ans :- DNS



Filter used: frame.number == 416

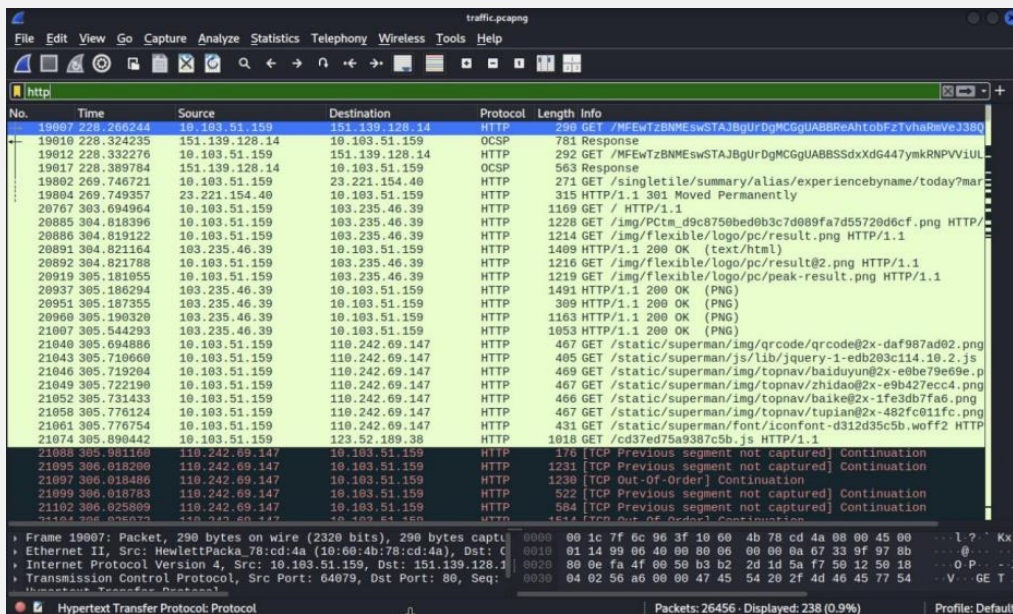
E12. Examine packet 416 in the network file and identify the destination IP address.

Ans :- 10.103.0.20



E13. Examine the network file and analyze the first HTTP packet; what is the source IP address?

Ans :- 10.103.51.159



E14. Examine the network file and analyze the first HTTP packet; what is the source port?

Ans :- 64079

No.	Time	Source	Destination	Protocol	Length	Info
18971	226.132586	10.103.51.159	147.75.38.124	TCP	55	[TCP Keep-Alive] 54166 → 443 [ACK] Seq=37194 Ack=67579 W
18974	226.178555	10.103.51.159	104.26.3.78	TCP	55	[TCP Keep-Alive] 65508 → 443 [ACK] Seq=31950 Ack=21769 W
18976	226.237897	104.26.3.78	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 65508 [ACK] Seq=21769 Ack=319
18977	226.263608	147.75.38.124	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54166 [ACK] Seq=67579 Ack=371
18978	226.289493	10.103.51.159	103.132.192.30	TCP	55	[TCP Keep-Alive] 64177 → 443 [ACK] Seq=22707 Ack=10010 W
18981	226.668412	103.132.192.30	10.103.51.159	TCP	60	[TCP Keep-Alive ACK] 443 → 64177 [ACK] Seq=10010 Ack=227
18985	227.032009	10.103.51.159	172.217.18.97	TCP	55	[TCP Keep-Alive] 50958 → 443 [ACK] Seq=1332 Ack=8825 Win
18986	227.087362	172.217.18.97	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 50958 [ACK] Seq=8825 Ack=1333
18988	227.251873	10.103.51.159	172.67.69.151	TCP	55	[TCP Keep-Alive] 54753 → 443 [ACK] Seq=1001 Ack=291791 W
18989	227.308660	172.67.69.151	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54753 [ACK] Seq=291791 Ack=16
19007	226.074554	10.103.51.159	213.150.150.183	TCP	66	[TCP Retransmission] 62448 → 443 [SYN] Seq=0 Win=0 Len=0
19004	226.211534	10.103.51.159	151.139.128.14	TCP	60	60 80 → 64079 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 S
19005	226.265841	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=14
19006	226.265945	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
19007	226.266244	10.103.51.159	151.139.128.14	HTTP	298	GET /MFewTzBNMEswSTA3BgUrDgMcGqUABBBReAhtobfzTvhARmVeJ380
19008	226.320002	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [ACK] Seq=1 Ack=237 Win=64128 Len=0
19009	226.323598	151.139.128.14	10.103.51.159	TCP	499	80 → 64079 [PSH, ACK] Seq=1 Ack=237 Win=64128 Len=445 [T
19010	226.324235	151.139.128.14	10.103.51.159	OCSF	781	Response
19011	226.324263	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=237 Ack=1173 Win=261376 Len=0
19012	226.332276	10.103.51.159	151.139.128.14	HTTP	292	GET /MFewTzBNMEswSTA3BgUrDgMcGqUABBBSSdxX0G447ymkRNPVv1UL
19013	226.343198	10.103.51.159	213.150.150.183	TCP	66	[TCP Retransmission] 60771 → 443 [SYN] Seq=0 Win=0 Len=0
19015	226.386257	151.139.128.14	10.103.51.159	TCP	60	80 → 64079 [ACK] Seq=1173 Ack=475 Win=64128 Len=0
19016	226.389657	151.139.128.14	10.103.51.159	TCP	499	80 → 64079 [PSH, ACK] Seq=1173 Ack=475 Win=64128 Len=445
19017	226.389784	151.139.128.14	10.103.51.159	OCSF	563	Response
19018	226.389815	10.103.51.159	151.139.128.14	TCP	54	64079 → 80 [ACK] Seq=475 Ack=2127 Win=262656 Len=0
19022	226.471182	10.103.51.159	52.178.182.73	TCP	66	50714 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
19024	226.492269	52.178.182.73	10.103.51.159	TCP	66	443 → 50714 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
19025	226.492338	10.103.51.159	52.178.182.73	TCP	54	50714 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
19026	226.492577	10.103.51.159	52.178.182.73	TLSv1.2	297	Client Hello [SNI=checkappexec.microsoft.com]

Filter used: http

E15. Analyze the network file and find the duration of the capture (in seconds).

Ans :- ≈ 348.11 seconds (about 5 min 48 sec)

E16. Analyze the network file and find the number of NBNS packets.

Ans :- 963 packets

No.	Time	Source	Destination	Protocol	Length	Info
18204	224.643010	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18915	224.915439	10.103.40.2	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18935	225.084585	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18953	225.369052	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<ic>
18955	225.404180	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18961	225.841794	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18967	226.048314	10.103.230.6	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18970	226.121700	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<ic>
18972	226.156093	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18980	226.594032	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18983	226.082326	10.103.230.6	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18984	226.873163	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<ic>
18995	227.560593	10.103.230.6	10.103.255.255	NBNS	92	Name query NB VUE<ic>
18996	227.788014	10.103.51.132	10.103.255.255	NBNS	92	Name query NB BOOMAIN<ic>
18998	227.822470	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<ic>
19040	226.506034	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<ic>
19048	226.713241	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19059	226.077881	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19063	226.351090	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<ic>
19065	226.469231	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19076	226.836049	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19091	230.225067	10.103.230.7	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19094	230.587966	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19117	231.673612	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<ic>
19122	232.091704	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19126	232.351443	10.103.230.10	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19127	232.433377	10.103.50.26	10.103.255.255	NBNS	92	Name query NB SETH-02<ic>
19129	232.551950	10.103.40.2	10.103.255.255	NBNS	92	Name query NB VUE<ic>
19132	232.844017	10.103.230.4	10.103.255.255	NBNS	92	Name query NB VUE<ic>

Filter used: nbns

E17. Analyze the network file and find the number of TCP packets sent with source port 443.

Ans :- 7,275 packets

No.	Time	Source	Destination	Protocol	Length	Info
18942	225.106375	142.250.185.238	10.103.51.159	TCP	60	443 → 59481 [ACK] Seq=7157 Ack=6855 Win=82176 Len=0
18945	225.106210	142.250.185.238	10.103.51.159	TLSv1.3	376	Application Data
18946	225.106324	142.250.185.238	10.103.51.159	TLSv1.3	377	Application Data, Application Data
18947	225.106324	142.250.185.238	10.103.51.159	TLSv1.3	93	Application Data
18951	225.176298	104.16.86.20	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 53514 [ACK] Seq=4223 Ack=1102
18952	225.223855	142.250.185.238	10.103.51.159	TCP	60	443 → 59481 [ACK] Seq=7841 Ack=6925 Win=82176 Len=0
18958	225.795847	109.206.188.82	10.103.51.159	TLSv1.3	310	Server Hello, Change Cipher Spec, Application Data, Appl
18962	225.858246	109.206.188.82	10.103.51.159	TCP	60	443 → 62549 [ACK] Seq=257 Ack=720 Win=65535 Len=0
18963	225.858432	109.206.188.82	10.103.51.159	TCP	60	443 → 62549 [ACK] Seq=257 Ack=1494 Win=65535 Len=0
18968	226.051985	185.172.90.251	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 56813 [ACK] Seq=22119 Ack=119
18976	226.237097	104.26.3.78	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 65508 [ACK] Seq=21769 Ack=319
18977	226.263608	147.75.38.124	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 54106 [ACK] Seq=67579 Ack=371
18981	226.608412	103.132.192.30	10.103.51.159	TCP	60	[TCP Keep-Alive ACK] 443 → 64177 [ACK] Seq=10010 Ack=227
18986	227.087362	172.217.18.97	10.103.51.159	TCP	66	[TCP Keep-Alive ACK] 443 → 50958 [ACK] Seq=8825 Ack=1333
18988	227.032000	172.67.69.151	10.103.51.159	TCP	60	443 → 54753 [ACK] Seq=1601 Ack=291791
19024	228.492269	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
19027	228.563628	109.206.188.82	10.103.51.159	TCP	60	443 → 51504 [RST, ACK] Seq=1 Ack=518 Win=65535 Len=0
19029	228.566680	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=1 Ack=244 Win=525312 Len=1460 [TCP
19031	228.568851	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=1461 Ack=244 Win=525312 Len=1460 [
19033	228.568964	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=2921 Ack=244 Win=525312 Len=1460 [
19035	228.569076	52.178.182.73	10.103.51.159	TCP	1514	443 → 58714 [ACK] Seq=4381 Ack=244 Win=525312 Len=1460 [
19037	228.569186	52.178.182.73	10.103.51.159	TLSv1.2	1291	Server Hello, Certificate, Certificate Status, Server Ke
19041	228.626677	109.206.188.82	10.103.51.159	TCP	60	443 → 57105 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1
19044	228.651282	52.178.182.73	10.103.51.159	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
19049	228.726471	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7129 Ack=2326 Win=525568 Len=0
19050	228.726689	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7129 Ack=5206 Win=525568 Len=0
19051	228.726942	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7129 Ack=8083 Win=525568 Len=0
19052	228.731116	52.178.182.73	10.103.51.159	TLSv1.2	636	Application Data
19057	228.805432	52.178.182.73	10.103.51.159	TCP	60	443 → 58714 [ACK] Seq=7712 Ack=8084 Win=525568 Len=0

Filter used: tcp.srcport == 443

E18. Examine the network file and find the number of TCP packets sent to destination port 443.

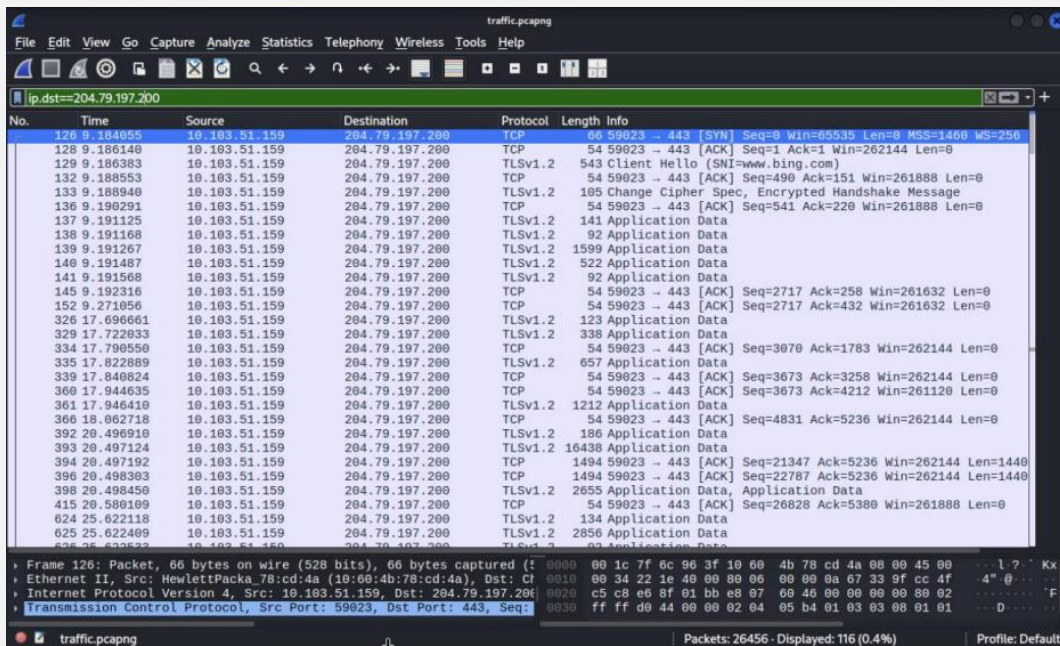
Ans :- 5,966 packets

No.	Time	Source	Destination	Protocol	Length	Info
18938	225.102926	10.103.51.159	142.250.185.238	TCP	84	59481 → 443 [ACK] Seq=6855 Ack=7157 Win=262912 Len=0
18940	225.103144	10.103.51.159	142.250.185.238	TLSv1.3	85	Application Data
18943	225.109622	10.103.51.159	104.16.86.20	TCP	55	[TCP Keep-Alive] 53514 → 443 [ACK] Seq=1101 Ack=4223 Win
18948	225.160352	10.103.51.159	142.250.185.238	TCP	54	59481 → 443 [ACK] Seq=6886 Ack=7841 Win=262400 Len=0
18949	225.161177	10.103.51.159	142.250.185.238	TLSv1.3	93	Application Data
18954	225.309620	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 69764 → 443 [SYN] Seq=9 Win=64240 L
18956	225.404979	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 83825 → 443 [SYN] Seq=9 Win=64240 L
18959	225.796281	10.103.51.159	109.206.188.82	TLSv1.3	134	Change Cipher Spec, Application Data
18960	225.796472	10.103.51.159	109.206.188.82	TLSv1.3	828	Application Data
18966	225.990664	10.103.51.159	185.172.90.251	TCP	55	[TCP Keep-Alive] 56813 → 443 [ACK] Seq=11932 Ack=22119 W
18971	226.132586	10.103.51.159	147.75.38.124	TCP	55	[TCP Keep-Alive] 54106 → 443 [ACK] Seq=37194 Ack=67579 W
18974	226.178555	10.103.51.159	104.26.3.78	TCP	55	[TCP Keep-Alive] 65508 → 443 [ACK] Seq=31950 Ack=21769 W
18978	226.280493	10.103.51.159	103.132.192.30	TCP	55	[TCP Keep-Alive] 64177 → 443 [ACK] Seq=22767 Ack=10010 W
18985	227.032009	10.103.51.159	172.217.18.97	TCP	55	[TCP Keep-Alive] 50958 → 443 [ACK] Seq=1332 Ack=8825 W
18988	227.251873	10.103.51.159	172.67.69.151	TCP	55	[TCP Keep-Alive] 54753 → 443 [ACK] Seq=1601 Ack=291791 W
18999	228.074354	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 62848 → 443 [SYN] Seq=0 Win=64240 L
19013	228.343196	10.103.51.159	213.155.156.183	TCP	66	[TCP Retransmission] 60771 → 443 [SYN] Seq=0 Win=64240 L
19022	228.417182	10.103.51.159	52.178.182.73	TCP	66	58714 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256
19025	228.492338	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
19026	228.492577	10.103.51.159	52.178.182.73	TLSv1.2	297	Client Hello (SNI=checkappexec.microsoft.com)
19028	228.564207	10.103.51.159	109.206.188.82	TCP	66	57105 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256
19030	228.568738	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=1461 Win=262144 Len=0
19032	228.568875	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=2921 Win=262144 Len=0
19034	228.568992	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=4381 Win=262144 Len=0
19036	228.569185	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=5841 Win=262144 Len=0
19038	228.569217	10.103.51.159	52.178.182.73	TCP	54	58714 → 443 [ACK] Seq=244 Ack=7078 Win=260664 Len=0
19039	228.575290	10.103.51.159	52.178.182.73	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Hand
19042	228.626144	10.103.51.159	109.206.188.82	TCP	54	57105 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
19043	228.626429	10.103.51.159	109.206.188.82	TLSv1	571	Client Hello (SNI=sync.evolution.ai)

Filter used: tcp.dstport == 443

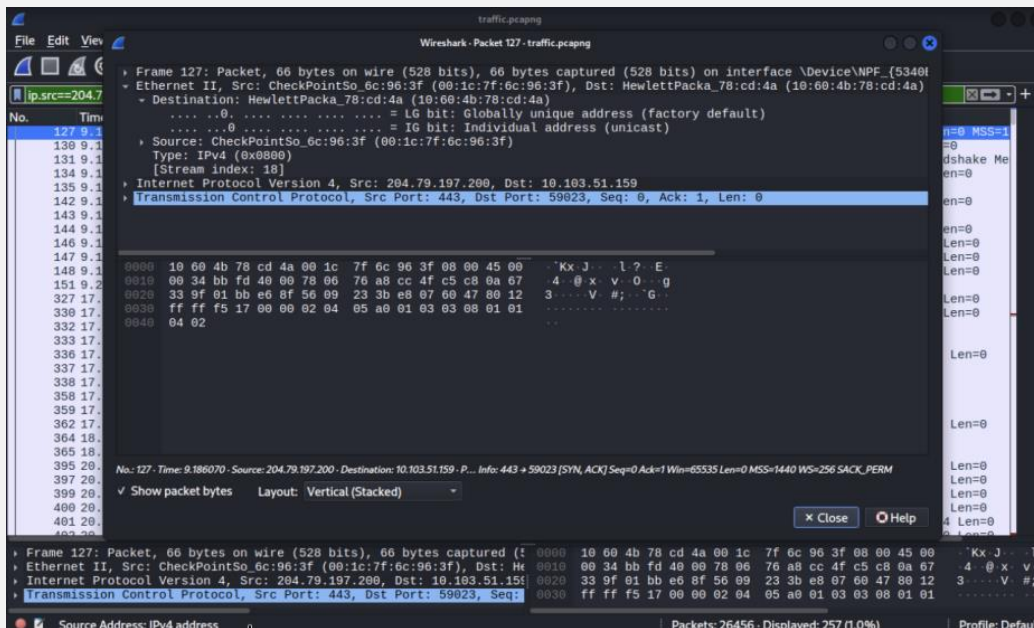
E19. Examine the network file; what is the number of packets sent to the destination IP address 204.79.197.200?

Ans :- 116 packets



E20. Analyze the network file and find the MAC address associated with the IP address 204.79.197.200. To which vendor does this MAC address belong?

Ans :- 00:1c:7f:6c:96:3f — Check Point Software Technologies



Observations

- Gateway / Firewall: MAC 00:1C:7F:6C:96:3F appears as the gateway MAC for all external IP communications, indicating a perimeter firewall/gateway on the segment.

- DNS Infrastructure: 10.103.0.20 functions as the internal DNS resolver — all DNS queries observed, including baidu.com and Microsoft domains, route through this host.
- Baidu Browsing Activity: Host 10.103.51.159 performed extensive browsing of www.baidu.com and related subdomains (image, ss.bdimg.com, hectorstatic.baidu.com, etc.).
- HTTPS Traffic Dominance: TCP port 443 accounts for 7,275 source-port and 5,966 destination-port packets, far exceeding plain HTTP (80) — consistent with a modern, encryption-aware environment.
- Windows Domain / NetBIOS Environment: 518 SMB/Browser broadcast packets and 963 NBNS packets indicate an active Windows LAN with multiple hosts performing NetBIOS name resolution and browser announcements.
- DHCP Infrastructure: Only 5 DHCP packets were captured, suggesting mostly stable IP leases during the ~348-second capture window, with one host releasing/renewing.

Conclusion

This analysis extracted all 20 required data points from the traffic.pcapng capture. Each answer above was derived directly from packet-level evidence (protocol headers and, where relevant, application-layer payloads) and is reproducible in Wireshark using the filters listed. The network under analysis is a Windows-based LAN behind a Check Point firewall/gateway, with active HTTPS browsing, SMB/NetBIOS name services, DNS resolution, and user activity directed toward both Microsoft and Baidu services.